

Problem 19

Determine the limit to which the following expression tends as $n \rightarrow \infty$:

$$a_n = \frac{1}{n} \sum_{k=1}^n \ln k.$$

Solution

Step 1: Recognize the Sum as a Logarithm of a Product

Using the property of logarithms that $\sum \ln k = \ln \prod k$:

$$\sum_{k=1}^n \ln k = \ln \left(\prod_{k=1}^n k \right) = \ln(n!).$$

Therefore:

$$a_n = \frac{1}{n} \ln(n!) = \frac{\ln(n!)}{n}.$$

Step 2: Apply Stirling's Approximation

Stirling's approximation states that:

$$\ln(n!) = n \ln n - n + \frac{1}{2} \ln(2\pi n) + O\left(\frac{1}{n}\right).$$

For large n , the dominant terms are:

$$\ln(n!) \sim n \ln n - n.$$

Step 3: Divide by n

Dividing by n :

$$\frac{\ln(n!)}{n} \sim \frac{n \ln n - n}{n} = \ln n - 1.$$

More precisely:

$$\frac{\ln(n!)}{n} = \ln n - 1 + \frac{\ln(2\pi n)}{2n} + O\left(\frac{1}{n^2}\right).$$

Step 4: Evaluate the Limit

As $n \rightarrow \infty$:

$$\frac{\ln(2\pi n)}{2n} \rightarrow 0 \quad (\text{since logarithmic growth is much slower than linear}).$$

Therefore:

$$\lim_{n \rightarrow \infty} \frac{\ln(n!)}{n} = \lim_{n \rightarrow \infty} \left(\ln n - 1 + \frac{\ln(2\pi n)}{2n} \right) = \lim_{n \rightarrow \infty} (\ln n - 1) = \infty.$$

Since $\ln n \rightarrow \infty$ as $n \rightarrow \infty$, we have:

$$\lim_{n \rightarrow \infty} a_n = \infty.$$

Step 5: Alternative Approach Using Riemann Sums

We can also interpret this limit using Riemann sums. Consider:

$$\frac{1}{n} \sum_{k=1}^n \ln k = \sum_{k=1}^n \ln k \cdot \frac{1}{n}.$$

This can be viewed as a Riemann sum for the integral:

$$\int_0^1 \ln(nx) dx,$$

where we partition $[0, 1]$ into n subintervals and sample at points $x_k = \frac{k}{n}$, giving $\ln(nx_k) = \ln k$.

However, this interpretation is subtle because the function depends on n . A better approach is to rewrite:

$$\frac{1}{n} \sum_{k=1}^n \ln k = \frac{1}{n} \sum_{k=1}^n \ln \left(\frac{k}{n} \cdot n \right) = \frac{1}{n} \sum_{k=1}^n \left[\ln \left(\frac{k}{n} \right) + \ln n \right].$$

This gives:

$$a_n = \frac{1}{n} \sum_{k=1}^n \ln \left(\frac{k}{n} \right) + \ln n.$$

The first term is a Riemann sum:

$$\frac{1}{n} \sum_{k=1}^n \ln \left(\frac{k}{n} \right) \rightarrow \int_0^1 \ln x dx.$$

Computing the integral:

$$\int_0^1 \ln x dx = [x \ln x - x]_0^1 = (1 \cdot 0 - 1) - \lim_{x \rightarrow 0^+} (x \ln x - x) = -1 - 0 = -1.$$

Therefore:

$$a_n \rightarrow -1 + \ln n \rightarrow \infty \quad \text{as } n \rightarrow \infty.$$

This confirms our result using Stirling's approximation.

Step 6: Growth Rate

More precisely, we can say:

$$a_n \sim \ln n - 1 \quad \text{as } n \rightarrow \infty.$$

The expression grows logarithmically, which means it increases without bound but very slowly.

Final Answer

The limit of the expression is:

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \ln k = \infty.$$

More specifically, the asymptotic behavior is:

$$a_n \sim \ln n - 1 \quad \text{as } n \rightarrow \infty.$$